

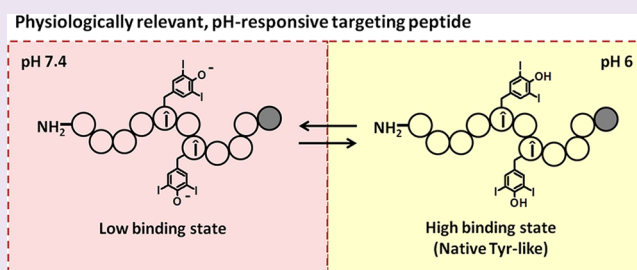
Reversibly Switchable, pH-Dependent Peptide Ligand Binding via 3,5-Diiodotyrosine Substitutions

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Supporting Information

ABSTRACT: Cell type-specific targeting ligands utilized in drug delivery applications typically recognize receptors that are overexpressed on the cells of interest. Nonetheless, these receptors may also be expressed, to varying extents, on off-target cells, contributing to unintended side effects. For the selectivity profile of targeting ligands in cancer therapy to be improved, stimuli-responsive masking of these ligands with acid-, redox-, or enzyme-cleavable molecules has been reported, whereby the targeting ligands are exposed in specific environments, e.g., acidic tumor hypoxia. One possible drawback of these systems lies in their one-time, permanent trigger, which enables the “demasked” ligands to bind off-target cells if released back into the systemic circulation. A promising strategy to address the aforementioned problem is to design ligands that show selective binding based on ionization state, which may be microenvironment-dependent. In this study, we report a systematic strategy to engineer low pH-selective targeting peptides using an M2 macrophage-targeting peptide (M2pep) as an example. 3,5-Diiodotyrosine mutagenesis into native tyrosine residues of M2pep confers pH-dependent binding behavior specific to acidic environment (pH 6) when the amino acid is protonated into the native tyrosine-like state. At physiological pH of 7.4, the hydroxyl group of 3,5-diiodotyrosine on the peptide is deprotonated leading to interruption of the peptide native binding property. Our engineered pH-responsive M2pep (Ac-Y-Î-Î) binds target M2 macrophages more selectively at pH 6 than at pH 7.4. In addition, 3,5-diiodotyrosine substitutions also improve serum stability of the peptide. Finally, we demonstrate pH-dependent reversibility in target binding via a postbinding peptide elution study. The strategy presented here should be applicable for engineering pH-dependent functionality of other targeting peptides with potential applications in physiology-dependent *in vivo* targeting applications (e.g., targeting hypoxic tumor/inflammation) or in *in vitro* receptor identification.



Targeting ligands are an important feature of macromolecular drug delivery carriers because biologics such as proteins or nucleic acids are too large for passive diffusion through the plasma membrane and rely on receptor-mediated endocytosis for internalization.^{1–3} As a result, preferential drug delivery to desired cells can also be achieved by modifying delivery formulations with targeting ligands that recognize receptors expressed on the cells of interest.⁴ In an effort to develop targeted drug delivery therapeutics for different diseases, a plethora of research has been carried out by many research groups to identify targeting ligands for a diverse array of target biomolecules or cell types.^{1,2}

A significant challenge in the development of targeted drug delivery carriers for cancer applications is the paucity of truly unique receptors for the target cells. Several tumor-targeting strategies utilize ligands for growth factor receptors (e.g., epidermal growth factor receptor (EGFR) and human epidermal growth factor receptor 2 (HER2)), folate/transferrin receptors, or RGD-binding integrins.^{5–8} Because these receptors are overexpressed, but not exclusively, on cancer cells, unintended binding could occur on noncancer cells.^{7,9,10}

Further complicating the design consideration, a dilemma exists where tumor heterogeneity and/or receptor shedding compromise targeting efficiency of highly tumor-specific ligands, whereas utilization of ligands recognizing more ubiquitously expressed receptors may incur greater off-target side effects.^{11–13} The latter may also be exacerbated by increased avidity when presented on drug carriers in a multivalent fashion. Thus, careful selection of ligands is paramount to the successful design of effective targeted drug delivery systems.^{1,4}

Inspired by the development of stimuli-responsive prodrugs such as those activated upon encountering enzymes, reactive oxygen species, or an acidic/reductive environment, the same design principle has been employed by several research groups to improve the selectivity of targeting ligands. As an example, Zhu et al. has reported the development of mannose-modified nanoparticles for targeting tumor-promoting, tumor-associated macrophages (TAMs), where selectivity over liver-resident

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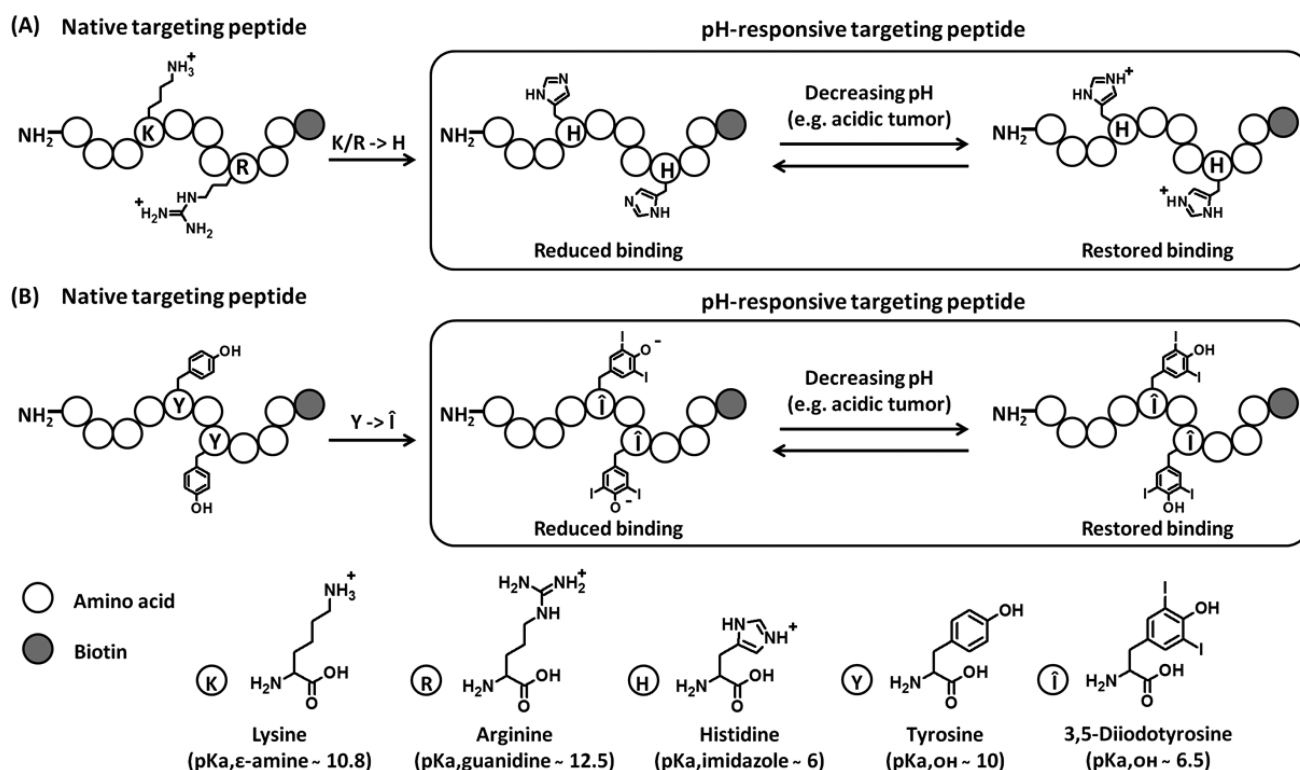


Figure 1. Theoretical illustration of tumor microenvironment (low pH)-responsive targeting peptide engineering via (A) histidine and (B) 3,5-diiodotyrosine substitution(s).

Table 1. Amino Acid Sequences and Molecular Weights of M2 Macrophage-Targeting Peptides

Peptide	Amino acid sequence	Calculated Mw (Da)	Measured Mw (Da)
M2pepBiotin	YEQDPWGVKWWYGGGSKK (K-Biotin)	2,524.86	2,524.21
AcM2pep(RY)Biotin	Ac-YEQDPWGVRYWYGGGSKK (K-Biotin)	2,700.06	2,699.45
M2pep(H9)Biotin	YEQDPWGVHWWYGGGSKK (K-Biotin)	2,662.00	2,661.95
I-W-Y	IYEQDPWGVRRWWYGGGSKK (K-Biotin)	2,932.85	2,931.71
Y-I-Y	YEQDPWGVRIWYGGGSKK (K-Biotin)	2,909.81	2,909.33
Y-W-I	YEQDPWGVRIWYGGGSKK (K-Biotin)	2,932.85	2,933.13
I-W-I	IYEQDPWGVRRWWYGGGSKK (K-Biotin)	3,184.65	3,184.41
Y-I-I	YEQDPWGVRIWYGGGSKK (K-Biotin)	3,161.61	3,161.65
Ac-Y-I-I	Ac-YEQDPWGVRIWYGGGSKK (K-Biotin)	3,203.61	3,204.08
Ac-Y-Y-F	Ac-YEQDPWGVRYWYGGGSKK (K-Biotin)	2,815.90	2,815.50

Kupffer cells and splenic macrophages also expressing mannose receptors was enhanced by masking the nanoparticles with poly(ethylene glycol) (PEG) that can be cleaved off in the acidic tumor microenvironment.¹⁴ Alternatively, Liu et al. utilized a legumain-specific peptide substrate to mask the cell-penetrating property of TAT peptide, the activity of which was restored upon encountering cancer cells and TAMs over-expressing catalytically active, plasma membrane-exposed legumain.¹⁵ Although target selectivity is promoted in these stimuli-activated targeting systems, one potential drawback lies in their nature of one-time activation, which may allow for off-target binding once the “demasked” ligand-decorated carriers transport back into systemic circulation.

One potential strategy to obtain reversibly activated targeting ligands is to develop pH-sensitive peptides that bind to their target based on their ionization state. With regard to biological applications in cancer, we hypothesized that acidic tumor microenvironment-selective targeting peptides could be engineered by replacing their critical binding residues with histidine (imidazole pK_a ~ 6) and/or 3,5-diiodotyrosine

(phenolic pK_a ~ 6.5¹⁶). Specifically, if the targeting peptides contain basic residues (lysine/arginine) whose cationic charges are involved in electrostatic interaction with negatively charged residues on their receptors, their corresponding substitution(s) with histidine could confer tumor microenvironment (low pH)-selective protonation of the substituted amino acids and, hence, restoration of target binding activity only in such a micro-environment (Figure 1A). Similarly, if the targeting peptides contain tyrosine residues whose hydroxyl groups are required for engagement with its receptor (e.g., through hydrogen bonding), their corresponding substitution(s) with 3,5-diiodotyrosine could confer an analogous pH-responsive binding behavior (Figure 1B).

As proof of feasibility, we investigate, in this study, the effect of histidine and 3,5-diiodotyrosine substitution(s) in an M2 macrophage-targeting peptide (M2pep) on pH-dependent binding activity.¹⁷ M2pep peptide has been previously shown to bind to M2 macrophages and M2-like TAMs, where the latter are also known to preferentially localize in the low pH, hypoxic region of tumors.^{17–20} In addition, the peptide

contains a known lysine essential for target binding as well as multiple tyrosine residues allowing for systematic substitution(s) and investigation.¹⁸ Specific to M2pep, we found that appropriate substitutions of tyrosine with 3,5-diiodotyrosine resulted in desirable low pH-selective binding activity of the engineered M2pep to target M2 macrophages. Because M2-like TAMs are a known target for cancer immunotherapy owing to their tumor-promoting roles (e.g., stimulation of cancer cell proliferation, immunosuppression, as well as promotion of angiogenesis and metastasis), as opposed to M1-like TAMs, which possess tumoricidal functions,²¹ the low pH-responsive M2pep reported here could be of translational significance in the development of M2-like TAM-targeted drug delivery systems. Further complementing M2-to-M1 selectivity of the original M2pep, a second layer of selectivity is encoded into the new peptide based on low pH activation.

RESULTS AND DISCUSSION

Peptide Synthesis. A series of peptides investigated in this study was successfully synthesized, purified, and identity-confirmed via matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-ToF MS) (Table 1). 3,5-Diiodotyrosine (**1**) was readily incorporated into the peptides using standard automatic peptide synthesis with no observable side reaction on the unprotected hydroxyl group of the amino acid. An initial attempt to acetylate the N-terminus of Y-**1**-**1** peptide with acetic anhydride resulted in over-acetylation at the hydroxyl groups of 3,5-diiodotyrosine. This issue was solved by coupling the last amino acid with Ac-Tyr(tbu)-OH.

pH-Dependent Binding Activity of M2pep and AcM2pep(RY) at pH 6 and 7.4. The original M2 macrophage-binding peptide (M2pepBiotin) and its affinity-enhanced analogue AcM2pep(RY)Biotin were first evaluated for their pH-dependent binding activities on M1 and M2 macrophages at pH 7.4 (physiological pH) and 6 (pathologically hypoxic pH). M1 macrophages were used as a control to evaluate selectivity of these peptides. Both M2pepBiotin and AcM2pep(RY)Biotin bind to M2 macrophages with good selectivity over M1 macrophages at both pH 6 and 7.4, whereas M2pepBiotin binds to M2 macrophages with higher affinity at pH 7.4 than at pH 6; the effect of pH on binding activity of AcM2pep(RY)Biotin is less consistent and appears to be concentration-dependent, slightly favoring binding at pH 6 as the peptide concentration is increased (Figure 2). For tumor microenvironment-targeting application, highly selective binding at a more acidic pH is preferred. Hence, these results motivated our effort to engineer new M2pep analogues that enhance high affinity binding to M2 macrophages at low pH.

The Effect of K9H Substitution on pH-Dependent Binding Activity. Histidine is a natural amino acid with an imidazole side chain that possesses a pK_a value around 6 and is variably ionized at physiological (7.4) versus endo/lysosomal (5–6) pH. Its modification is widely used in protein engineering to confer physiologically relevant, pH-dependent functionality to proteins/antibodies.^{22,23} With regard to peptide development, neonatal Fc receptor (FcRn)-binding peptides have been reported with endowed pH-dependent binding activity through the presence of histidine in the sequence.²⁴ We therefore investigated if a lysine to histidine substitution at the 9th position of M2pepBiotin (K9H) might enable low pH-selective binding activity of the peptide. The lysine at this position is known to influence binding activity of the peptide

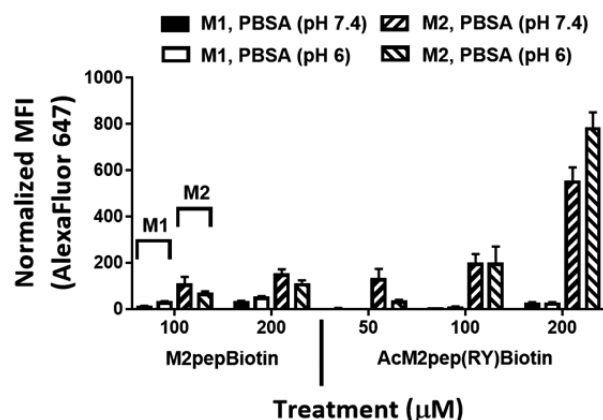


Figure 2. Binding M2pepBiotin and AcM2pep(RY)Biotin with M1 and M2 macrophages at pH 7.4 and 6.

because its substitution with arginine has been previously shown to improve affinity of the peptide.¹⁸ We hypothesize that pH-dependent, neutral-to-positive ionization of histidine at this position may result in pH-dependent binding behavior. Indeed, a reverse trend in binding activity of M2pep(H9)Biotin on M2 macrophages was observed where the peptide binds more strongly at pH 6 than at 7.4 (Figure 3). Nonetheless, the

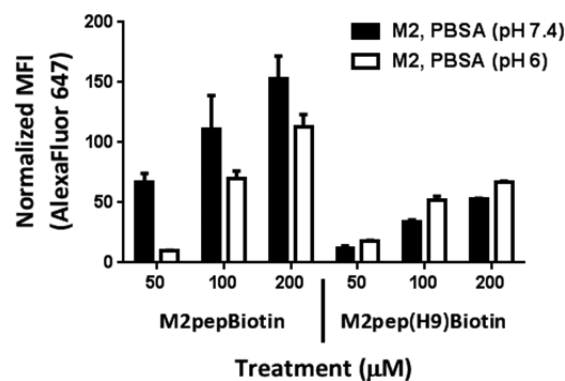


Figure 3. Binding of M2pepBiotin and M2pep(H9)Biotin with M2 macrophages at pH 7.4 and 6.

difference in binding activities at these pH values is relatively marginal, and the overall binding affinity of the peptide is reduced compared to that of the original M2pepBiotin. The reduction in binding affinity associated with the K9H modification may be due to the lower extent of protonation of histidine relative to that of lysine in this pH range or due to less favorable conformational fit of the peptide to its receptor owing to the presence of a bulky imidazole group. Although it is possible to investigate the effect of K9 substitution with 2,3-diaminopropionic acid (Dap), which is more structurally similar to lysine and possesses lower pK_a on its amino side chain (6.3–7.49 depending on the surrounding environment²⁵), we decided to keep the affinity-enhanced K9R modification and looked into modifications at other amino acid positions utilizing other non-natural amino acids.

3,5-Diiodotyrosine Substitutions Impart pH-Dependent Binding to M2pep. 3,5-Diiodotyrosine was chosen for investigation next because its phenolate group possesses a desirable pK_a around 6.5, and its Fmoc-protected amino acid is commercially available, allowing for incorporation into M2pep peptides via standard Fmoc/tbu solid-phase peptide synthesis.

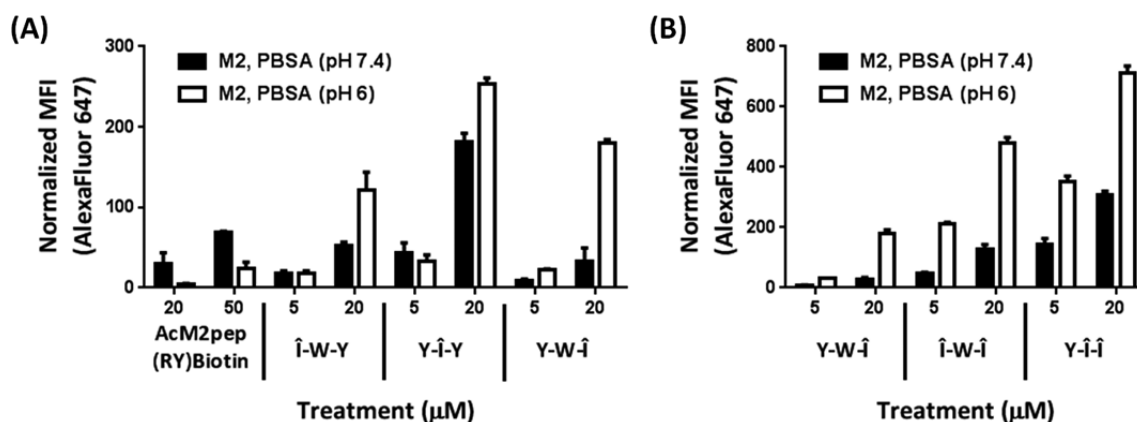


Figure 4. Binding of (A) monosubstituted and (B) disubstituted M2pep(R9) analogues with M2 macrophages at pH 7.4 and 6.

The original M2pepBiotin contains two tyrosines at the 1st and 12th positions, whereas AcM2pep(RY)Biotin contains an additional tyrosine at the 10th position, giving three possible sites for modification. M2pep(R9) analogues with a single 3,5-diiodotyrosine mutation at each tyrosine position (Î-W-Y, Y-Î-Y, and Y-W-Î) were synthesized and evaluated for pH-dependent binding activity on M2 macrophages. At 20 µM peptide treatment, all monosubstituted analogues show higher binding activity to M2 macrophages at pH 6 than at 7.4 with the Y-W-Î analogue (Y12Î substitution) having the highest low-pH selectivity (Figure 4A). Interestingly, all monosubstituted analogues also exhibit enhanced binding activity relative to AcM2pep(RY)Biotin.

Given the encouraging result of the monosubstituted analogues, we next studied the effect of dual 3,5-diiodotyrosine substitutions on binding affinity and pH selectivity of M2pep(R9). The most pH-selective modification (Y12Î) was preserved, whereas another 3,5-diiodotyrosine substitution was added at either the 1st (Î-W-Î) or 10th (Y-Î-Î) position. Both disubstituted analogues show further enhancement in binding activity over the monosubstituted ones while also retaining good low-pH selectivity (Figure 4B). Toxicity of these peptides was observed at higher concentrations, preventing us from accurately constructing the binding curves to determine $K_{D,7.4}/K_{D,6}$ for quantitative comparison of pH selectivity of each analogue (data not shown). Of note, the disubstituted analogues are also more toxic than the monosubstituted ones at higher concentrations, and hence, the trisubstituted analogue was not further investigated. Nonetheless, by systematically mutating different tyrosines in M2pep(R9) with 3,5-diiodotyrosine, we were able to engineer the low pH-selective binding property into the M2pep(R9) peptide and obtain Y-Î-Î as the lead analogue, which possesses both enhanced M2 macrophage-binding affinity and low pH binding selectivity with the potential to improve serum stability as investigated in the next section. Although substituted tyrosines with tunable pK_a values have been previously used to study tyrosine-dependent activity of biological enzymes (e.g., oxidase activity of engineered myoglobin^{26,27}) as well as employed in tuning the range of pH sensitivity of membrane-disrupting peptides (e.g., diiodotyrosine-substituted GALA²⁸ and tetrafluorotyrosine-substituted magainin 2²⁹), their utility in engineering pH-responsive targeting ligands has, to our knowledge, not been reported to date. As demonstrated here, both histidine and 3,5-diiodotyrosine provide complementary options in optimization of physiologically relevant, pH-dependent functionality of peptides where

the former controls a neutral-to-positive switch and the latter controls a negative-to-neutral switch. In addition, this concept should also be applicable to other non-natural amino acids (e.g., Dap and tetrafluorotyrosine) whose side chain pK_a values lie in the physiologically relevant pH, further expanding the optimization toolbox with regard to both affinity and pH-sensitivity engineering.

Acetylated Y-Î-Î Peptide Exhibits Improved Serum Stability Compared to That of AcM2pep(RY)Biotin. N-terminal amines are known to have lower pK_a relative to the epsilon amines of lysine side chains and may contribute to pH-dependent functionality of peptides/proteins at the physiologically relevant pH.^{30,31} To investigate if a free N-terminal amine has an effect on pH-dependent binding activity of M2pep, we synthesized and compared the binding activities of Y-Î-Î and Ac-Y-Î-Î. No noticeable difference in binding activity of these peptides on M1 and M2 macrophages was observed at either pH (Figure 5A), implying that the N-terminal amine is not involved in the observed pH-dependent binding effect. Binding studies of the peptides were also performed on M1 macrophages to evaluate M2-to-M1 selectivity. For both peptides, an increase in binding activity on M1 macrophages is observed at pH 6, but the extent of binding remains substantially lower compared to that on M2 macrophages. With regard to tumor physiology, the hypoxic region of the tumor is known to more readily harbor TAMs of M2-like phenotype.^{19,20} Hence, some increase in binding activity of Y-Î-Î and Ac-Y-Î-Î on M1 macrophages at low pH is expected to minimally compromise their M2-to-M1 selectivity in the hypoxic tumor microenvironment. This development exemplifies how tumor physiological/biological context may be rationally utilized to guide the design and evaluation of targeting peptides.

Because N-terminal acetylation does not have an effect on the binding activity of Y-Î-Î, we next evaluated the serum stability of Ac-Y-Î-Î. Pleasingly, 3,5-diiodotyrosine substitutions in Ac-Y-Î-Î result in the delayed degradation kinetics of the peptide in serum compared to that in AcM2pep(RY)Biotin (Figure 5B). The intact Ac-Y-Î-Î was detectable under MALDI-ToF MS after serum incubation for at least 24 h, whereas AcM2pep(RY)Biotin was not. Following the serum incubation for 48 h, the degradation of Ac-Y-Î-Î was eventually observed at the Î10/W11 site, which coincides with the degradation site (Y10/W11) observed on AcM2pep(RY)Biotin at the 24 h time point. As demonstrated here, modifications of peptides with non-natural 3,5-diiodotyrosine at the protease-susceptible site may serve as a useful strategy to both confer pH-dependent

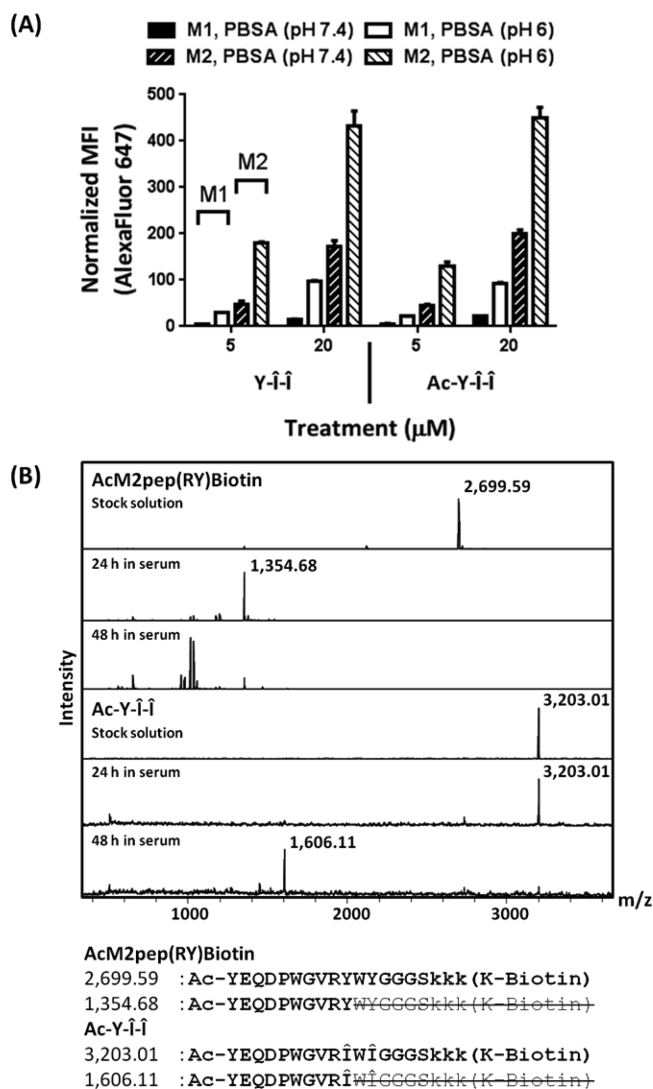


Figure 5. (A) Binding of Y-I-I and Ac-Y-I-I with M1 and M2 macrophages at pH 7.4 and 6. (B) Serum stability of AcM2pep(RY)-Biotin and Ac-Y-I-I in normal mouse serum.

functionality and retardation of peptide degradation kinetics in serum.

Tetrafluorophenol Moiety as an Alternative to 3,5-Diiodotyrosine Confirms the Role of Ionizable Tyrosine on pH-Dependent Binding Activity. To further confirm the role of ionizable tyrosine on pH-dependent binding activity of M2pep, we synthesized Ac-Y-Y-F, where the last tyrosine residue was replaced with 2,3,5,6-tetrafluoro-4-hydroxybenzoic acid-functionalized 2,3-diaminopropionic acid (F) (Figure 6A). A pK_a value of F on Ac-Y-Y-F was estimated from an experimental acid titration curve and was found to be comparable to I on Ac-Y-I-I (7.5–8) (Figure S1). As similarly observed in the binding data of Ac-Y-Y-I, Ac-Y-Y-F binds to M2 macrophages more strongly at pH 6 compared to its binding activity at pH 7.4 (Figure 6B). In addition, Ac-Y-Y-F also exhibits enhanced binding affinity over AcM2pep(RY)-Biotin. Because substitution of the original tyrosine on M2pep with either I or F may alter target binding affinity or specificity, we performed a competition binding assay and confirmed that both Ac-Y-Y-F and Ac-Y-I-I can compete with bound M2pep-displayed phage on M2 macrophages (Figure S2). Confirming

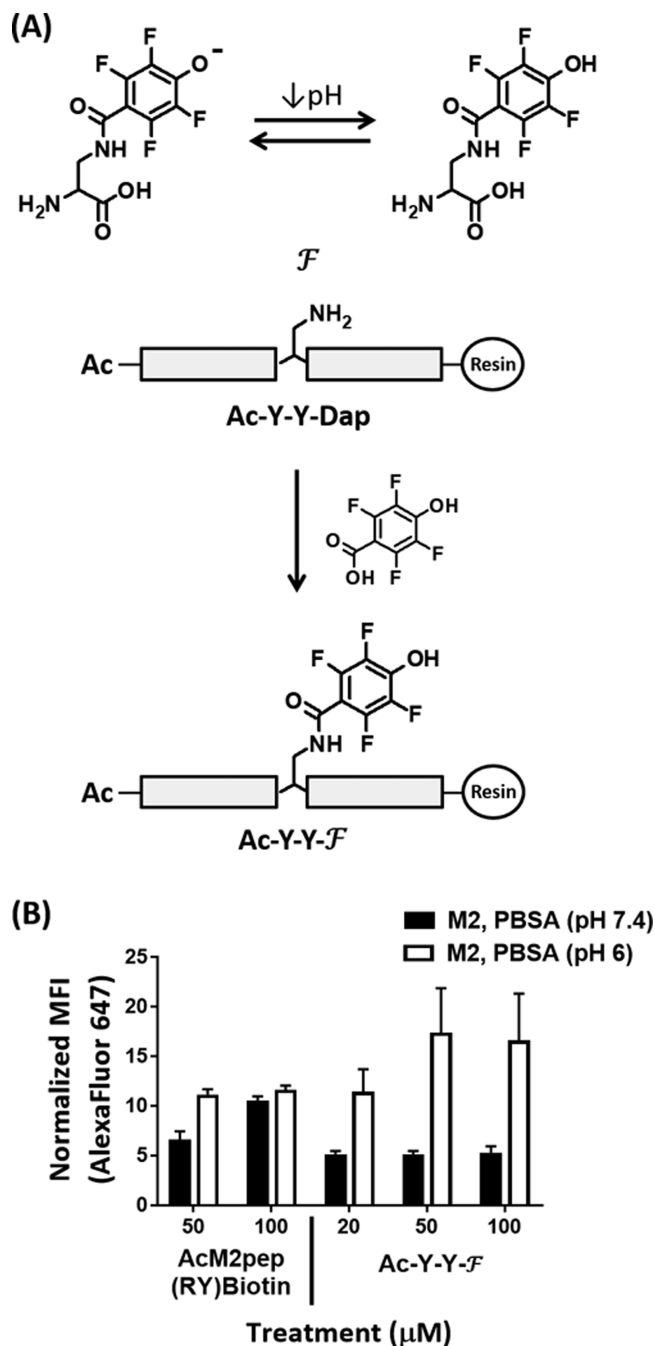


Figure 6. (A) Schematic of F and Ac-Y-Y-F on resin. (B) Binding of AcM2pep(RY)Biotin and Ac-Y-Y-F with M2 macrophages at pH 7.4 and 6.

with two physiologically ionizable tyrosine analogues (F and I), we highlight their role in augmenting low pH-selective binding of M2pep, the concept of which should be applicable to other peptides. Moreover, the pendant fluorine or iodine on these tyrosine analogues may further fine-tune binding activity of the modified peptides as seen with M2pep modifications.

Demonstration of Reversible, pH-Dependent Modulation of Peptide Binding Affinity. Finally, we envision that engineering pH-dependent functionality of peptides could be a useful strategy not only in the development of physiology-selective targeting peptides for *in vivo* targeting applications but also in *in vitro* experiments, e.g., pull-down assays for receptor identification. For the latter application, an experiment is

typically performed where a tag-labeled (e.g., biotin) targeting ligand is incubated with cell lysate containing its unknown receptor and is passed through the capture (e.g., neutravidin) beads where, ideally, only the receptor-bound ligand would be captured on the beads and subsequently eluted for analysis via sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) and liquid chromatography-tandem mass spectrometry (LC-MS/MS). Nonetheless, nonspecific adsorption of nontarget proteins/oligonucleotides is oftentimes observed resulting in false-positive results.³² In such a case, a pH-dependent receptor elution, as an alternative to traditional bead boiling, could therefore be a complementary tool in enabling more selective elution of target receptors. To test the pH-reversible binding behavior of our engineered pH-responsive Ac-Y-Î-Î, we performed a binding study of the peptide in its high binding buffer (PBSA, pH 6) on M2 macrophages followed by post-treatment incubation with PBSA at different pH levels (6, 7.4, and 8.5) (Figure 7A). In theory, post-

peptides as demonstrated here, may provide an additional strategy for this application.

CONCLUSIONS

We report here a systematic approach in engineering pH-dependent targeting functionality of M2 macrophage-targeting peptides utilizing physiologically relevant, ionizable amino acids (histidine, 3,5-diiodotyrosine). The strategy should be widely applicable in engineering pH selectivity of other targeting peptides with potential utility in targeting low pH, hypoxic disease environments such as in cancer or inflammation as well as in improving the protocol of *in vitro* experiments such as pull-down assays.

MATERIALS AND METHODS

Materials. Protected amino acids and 2-(6-chloro-1H-benzotriazole-1-yl)-1,1,3,3-tetramethylammonium hexafluorophosphate (HCTU) were purchased from AnaSpec (Fremont, CA) and AAPPTec (Louisville, KY). Fmoc-3,5-diiodo-Tyr-OH was purchased from Bachem (Torrance, CA). Ac-Tyr(tbu)-OH was purchased from Chem-Impex International (Wood Dale, IL). NovaPEG rink amide resin was purchased from EMD Millipore (Billerica, Massachusetts). Triisopropylsilane (TIPS), 1,3-dimethoxybenzene (DMB), N-methylmorpholine, piperidine, and 2,3,5,6-tetrafluoro-4-hydroxybenzoic acid were purchased from Sigma-Aldrich (St. Louis, MO). The Quant*Tag Biotin quantification kit was purchased from Vector Laboratories (Burlingame, CA). Streptavidin-Alexa Fluor 647 was purchased from BioLegend (San Diego, CA). Normal mouse serum (catalog no. 10410) was purchased from Thermo Fisher Scientific (Waltham, MA).

Peptide Synthesis. Biotinylated resin was synthesized following the previously described protocol.¹⁸ In brief, Fmoc-Lys(Mtt)-OH (4 equiv) was manually coupled to NovaPEG rink amide resin using HCTU (3.9 equiv) as an activator dissolved in 0.4 M N-methylmorpholine in DMF. The Mtt protecting group was removed by incubations in 2% TFA in DCM (10–15 times), and the epsilon amine was then coupled with D-biotin in the same manner. The biotinylated resin was used in the synthesis of all biotinylated peptides via a PS3 automated peptide synthesizer (Protein Technologies, Phoenix, AZ) following a standard Fmoc/tbu solid phase peptide synthesis with HCTU as an activator and piperidine as an Fmoc-deprotection agent. Fmoc-3,5-diiodo-Tyr-OH was used in the synthesis of 3,5-diiodotyrosine-containing peptide analogues. Acetylation of AcM2pep(RY)Biotin was performed on resin by incubation in acetic anhydride/TEA/DCM (0.5:0.5:5 v/v/v) for 2 h. N-terminal acetylation of Ac-Y-Î-Î was instead achieved by coupling with Ac-Tyr(tbu)-OH for the last amino acid residue. For the synthesis of Ac-Y-Y-F, the base peptide was synthesized on resin with Dap(Alloc) replacing the last tyrosine residue. The Alloc group was selectively deprotected by incubating the resin in a solution of Pd(PPh₃)₄ (0.2 equiv) and phenylsilane (20 equiv) in DCM for 20 min twice under nitrogen. The resin was washed with 5 mM sodium diethyldithiocarbamate solution in DMF and then coupled with 2,3,5,6-tetrafluoro-4-hydroxybenzoic acid following the same amino acid coupling procedure. All peptides were cleaved by incubation in TFA/TIPS/DMB (92.5:2.5:5 v/v/v) for 2.5 h followed by precipitation in cold ether twice. The crude peptides were purified on an Agilent 1200 high-performance liquid chromatography (HPLC) system (Agilent Technologies, Santa Clara, CA) using a Phenomenex Fusion-RP C18 semipreparative column (Phenomenex, Torrance, CA) with water (0.1% TFA) and ACN (0.1% TFA) as mobile phases. Molecular weights of the synthesized peptides were confirmed by MALDI-ToF MS.

Bone Marrow Harvest and Culture of Primary Macrophages.

All animal handling protocols were approved by the University of Washington Institutional Animal Care and Use Committee (IACUC). Bone marrow-derived cells were harvested from female c57bl/6 mice as previously described.¹⁸ In brief, femurs and tibia were excised from euthanized mice, nicked at each end, and flushed with RPMI 1640

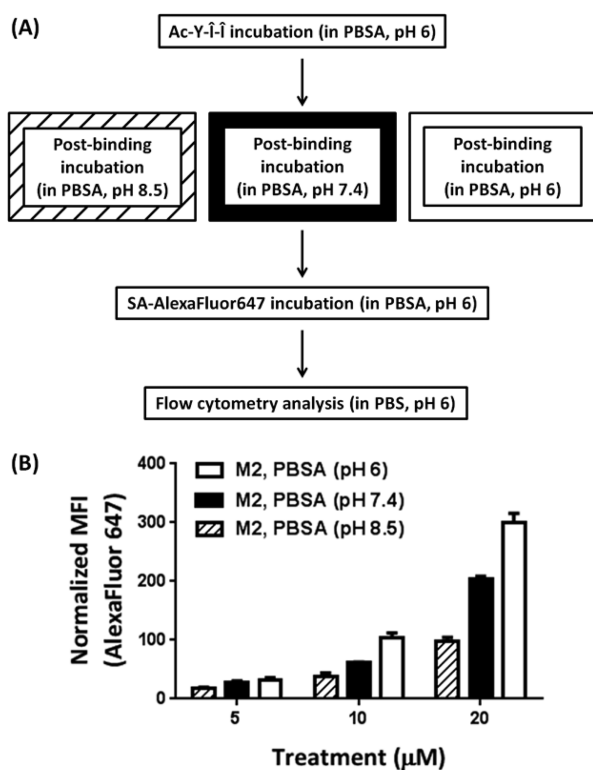


Figure 7. (A) Experimental schematic and (B) result of postbinding peptide elution study on M2 macrophages.

treatment incubation in higher pH buffer should deprotonate 3,5-diiodotyrosine on Ac-Y-Î-Î, ripping off its association with its target receptor. Indeed, we observed a reduction in median fluorescence intensity read-out as a function of increasing pH, which indicates the pH-dependent elution of the bound peptides off the target (Figure 7B). In a pull-down assay, although it may be possible to selectively elute target receptor by incubation in an excess amount of untagged ligand, enhanced avidity from the multivalent effect of the ligand-bound beads sometimes requires a high concentration of the untagged ligand for elution, which may not be achievable due to its solubility limit. Hence, pH-dependent elution, made possible by engineering a pH-dependent binding property of targeting

medium to obtain the cells. These cells were differentiated into macrophages by culturing on Petri dishes in RPMI 1640 supplemented with 20% donor horse serum (DHS), 1% antibiotic-antimycotic (AbAm), and 20 ng/mL of M-CSF. After 7 days of culture, macrophage activation was performed by replacing new media substituting M-CSF with either 25 ng/mL of IFN- γ and 100 ng/mL of LPS for M1 polarization or 25 ng/mL of IL-4 for M2 polarization. For M1 polarization, AbAm was replaced with 1% penicillin-streptomycin (Pen-Strep). The macrophages were maintained in the activation media for 2 days before use in binding studies.

In Vitro Binding Study. Stock peptide solutions (10 mg mL⁻¹ in water) were quantified by the Quant⁺Tag Biotin quantification kit and diluted with PBS (1% albumin) (PBSA) at pH 6 or 7.4 for binding studies. Macrophages were scraped off the Petri dishes and seeded on round-bottom 96-well plates (50,000 cells/well). The seeded macrophages were first equilibrated by washing with PBSA at the pH of interest (pH 6 or 7.4) before incubation in peptide solutions at the respective pH for 20 min on ice. Subsequently, the macrophages were washed twice in PBSA and probed for binding by incubation in streptavidin-Alexa Fluor 647 (1:1000 dilution) for 15 min on ice. After two more washes in PBSA, the macrophages were resuspended in PBS for analysis via MACSQuant Flow Cytometer (Miltenyi Biotec, San Diego, CA) or an Attune NxT Flow Cytometer (Thermo Fisher Scientific, Waltham, MA). Propidium iodide was added prior to running of each sample to assess for cell viability. All washing steps and incubations were performed in buffers at the desired pH under investigation (pH 6 or 7.4). Binding studies were performed with two samples per treatment, and the observed trends in binding were confirmed in at least two independent studies. For the postbinding peptide elution study, all treatment and washing steps were in PBSA at pH 6. In the study, an additional incubation in PBSA at pH 6, 7.4, or 8.5 for 20 min on ice was included following the peptide incubation step.

Serum Stability Study. The stock peptide solutions (30 μ L) were diluted into normal mouse serum (300 μ L). At different time points, an aliquot (40 μ L) was drawn, mixed with an equal volume of ACN to precipitate serum proteins, and centrifuged at 10,000 rpm for 5 min. The supernatant solution was transferred to a clean Eppendorf tube, and residual peptides in the pellet were further extracted by sonication in 1:1 H₂O/ACN (80 μ L) for 10 min. The sonicated mixture was centrifuged, and the new supernatant solution was pooled with the former and dried on a SpeedVac machine. The peptide extract was resuspended by sonication for 10 min in H₂O (50 μ L) followed by centrifugation. The supernatant was analyzed for peptide degradation pattern via MALDI-ToF MS.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscchembio.8b00171.

Titration curves of Ac-Y-Y-F and Ac-Y-I-I and competition binding study of M2pep-displayed M13 phages with M2pep analogues (PDF)

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Notes

The authors declare no competing financial interest.

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